

ETHICAL CONSIDERATIONS

AI TRIAGE SUPPORT SYSTEM CO-DESIGN CANVAS

Consider potential ethical problems, and potential solutions –both from the user's and robot's perspectives.
Consider the boxes to be guidelines: you don't need to fill each one.

Physical safety

Machines can pinch or crush the user. How is this mitigated?

PROBLEM	USER		SOLUTION
	ROBOT		
- Over-reliance on AI recommendations could delay recognition of critical patients. - Incorrect recommendations may influence clinical judgement if accepted without review.		- Nurse remains the final decision-maker. - Clear override button with mandatory reason logging. - AI presented as <u>decision support only</u>, not autonomous decision-making. - Low-confidence recommendations prompt reassessment rather than automatic acceptance.	

Data security

Is the robot in a unique data collection position? How is the user's data protected?

PROBLEM	USER		SOLUTION
	ROBOT		
- Patient information contains sensitive personal and clinical data. - Patient portal sessions expire automatically after the visit or period of inactivity. - Risk of unauthorised access or accidental disclosure.		- Secure authentication. - Encrypted storage and transmission. - Role-based access control. - Complete audit logs.	

Transparency

How does the robot share an accurate perception of its abilities, intentions and constraints, so the user can evaluate their trust in it?

PROBLEM	USER		SOLUTION
	ROBOT		
- Nurses may not understand why the AI suggested a particular ESI level. - Black-box predictions reduce trust.		- Display confidence score. - Show key contributing clinical features. - Clearly indicate that recommendations are advisory. - Patient-facing screens clearly explain that AI does not replace clinical staff.	

Equality across users

Robots' algorithms can be biased. A robot's appearance could reinforce harmful stereotypes. What are potential issues?

PROBLEM	USER		SOLUTION
	ROBOT		
- Performance may vary across different patient groups. - Training data may contain bias.		- Regular fairness audits. - Continuous monitoring. - Diverse validation datasets. - Human review of unusual cases. - Provide accessible interfaces, multilingual support and hospital-issued devices for patients who cannot use smartphones.	

Emotional consideration

People have been shown to form emotional attachments to robots, as if they were alive. Is this a potential problem?

PROBLEM	USER		SOLUTION
	ROBOT		
- Staff may either distrust the AI or rely on it too heavily. - Long waiting times and unclear communication may increase anxiety for patients.		- Simple interface. - Training before deployment. - Keep nurse in control. - Minimise unnecessary alerts. - Provide patients with clear status updates and instructions while waiting.	

Behaviour enforcement

People could transfer their inappropriate behaviour, such as rudeness, from robots to humans. How is this mitigated?

PROBLEM	USER		SOLUTION
	ROBOT		
- Users may ignore alerts or misuse overrides. - Frequent alerts may contribute to alarm fatigue.		- Override reasons recorded. - High-risk alerts prioritised. - Regular review of override trends. - Optimise alert thresholds. - Patient-submitted symptom updates require nurse acknowledgement before being closed.	